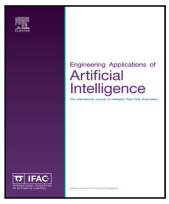
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Research paper



An efficient knowledge tracing model via Mamba Contextual Encoding and Dynamic Sparse Attention mechanism

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ABSTRACT

Knowledge Tracing (KT) predicts learners' future performance by analyzing their historical learning records. While deep learning-based knowledge tracing models have significantly improved prediction performance, they suffer from substantial computational overhead and inefficiency when handling long interaction sequences. To address this problem, we propose an efficient knowledge tracing model named **M**amba **C**ontextual Encoding and Dynamic **S**parse Attention Mechanism-based **K**nowledge **T**racing (MCSKT). Firstly, leveraging the selective state space structure and linear-time complexity of Mamba, we design a dual-encoder composed of a question encoder and a knowledge encoder, which structurally disentangles the contextual dependencies at the question level and concept level. This design enhances semantic modeling capability while maintaining computational efficiency. Secondly, we propose a dynamic k-sparse attention mechanism to overcome the adaptability constraints inherent in traditional sparse attention methods that rely on manually configured static thresholds. This novel mechanism dynamically adjusts the filtering range of historical interactions, adaptively balancing noise suppression and critical information retention, while significantly reducing computational complexity. Experimental results demonstrate that MCSKT achieves an average improvement of 3.7% in Area Under the Curve (AUC) and 2.9% in Accuracy (ACC) across four public datasets. Moreover, compared with the state-of-the-art model, MCSKT achieves substantial acceleration, running approximately 10.1 times faster during training and 3.5 times faster during inference. In addition, the growth rate of time consumption for MCSKT is markedly slower than that of competing models as sequence length increases, highlighting its advantage in processing long-sequence data.

1. Introduction

Knowledge Tracing (KT) was first introduced by Corbett and Anderson in 1994 (Corbett and Anderson, 1994a) with the goal of modeling a learner's mastery of knowledge concepts by analyzing their historical learning records, and subsequently predicting future performance (Wang et al., 2023b; Sun et al., 2022). With the rapid development of artificial intelligence and smart education, learning activities have increasingly shifted from traditional classrooms to online educational platforms such as MOOCs and Intelligent Tutoring Systems (ITS) (Sun et al., 2024). These platforms have accumulated massive amounts of learning behavior data, including questions, knowledge concepts, and responses (Qiu et al., 2023; Cui et al., 2023a), thus providing a rich foundation for KT research. As illustrated in Fig. 1, the scale of KT-related datasets has grown exponentially in recent years. For example, the EdNet dataset released in 2017 contains roughly 100 times more learners than the 2005 Algebra dataset and about 400 times more interactions than the 2011 Statics dataset (Cui et al., 2023b; Cao and Zhang, 2024). While such large-scale data has facilitated the development of Deep Learning-based Knowledge Tracing (DLKT), enabling models to better capture complex learning behavior patterns, it has also substantially increased the computational cost of training and inference. Therefore, achieving efficient knowledge tracing under large-scale online learning environments has become a key challenge and the primary focus of DLKT.

Existing DLKT models can be categorized into two main types based on their core sequence modeling mechanisms: Recurrent Neural Network (RNN)-based models and Attention-based models, most of which universally take improving prediction accuracy as their primary objective (Cao and Zhang, 2024). However, with the continued growth in data size, DLKT models face a dilemma between efficiency and prediction performance: the improvement of accuracy is often accompanied by structural complexity, which in turn intensifies the consumption of

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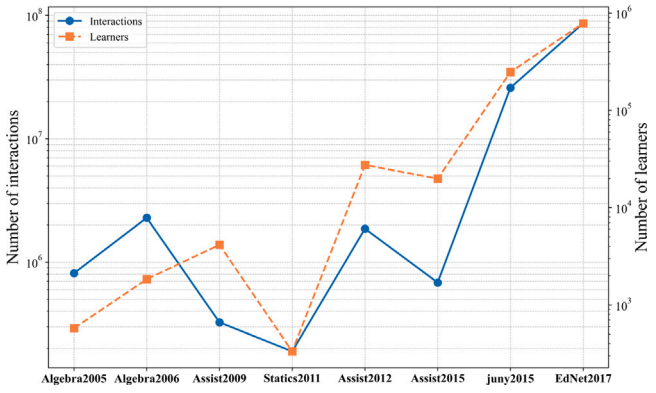


Fig. 1. Comparison of dataset scales across different knowledge tracing benchmarks. The rapid growth in the number of learners and interactions highlights computational efficiency as a critical challenge for models deployed in large-scale online education scenarios.

computational resources. RNN-based DLKT models effectively capture temporal dependencies but suffer from limited training parallelism due to their inherently sequential structure(Cao and Zhang, 2024). In contrast, attention-based DLKT models achieve higher prediction performance and support parallel computation, yet their reliance on global attention results in high computational complexity $O(N^2)$ and significant resource consumption. Therefore, the balance between prediction performance improvement and computational overhead reduction is a key challenge in current DLKT research. Cao et al. first introduced Mamba into the field of KT and proposed the Mamba4KT model (Cao and Zhang, 2024), which prioritizes model efficiency and resource utilization. This model not only ensures prediction accuracy but also achieves a balance between time and space consumption, providing an important approach to address these issues. Although the Mamba inherently benefits from linear-time complexity, its original architecture is designed for general sequence modeling and cannot be directly applied to KT tasks without structural adaptation.

To this end, we reconstruct the task-level architecture based on the Mamba framework and propose an efficient KT model named MCSKT. MCSKT consists of two core modules: a context-aware representation module and a knowledge state extraction module. Firstly, to address the limitation of traditional models in jointly modeling question-level semantics and concept-level dependencies, we develop a Mamba-based context-aware representation module. This module comprises a question encoder (Q-encoder) and a knowledge encoder (K-encoder). The Q-encoder captures semantic relationships among questions, while the K-encoder extracts conceptual cognitive dependencies from the response sequence. By structurally disentangling question-level and concept-level contextual information, this module substantially enhances the model's capacity to understand learning behaviors. Secondly, we design a knowledge state extraction module to more accurately infer learners' knowledge states. This module introduces sparse attention to focus on key historical interactions and further incorporates a dynamic k-sparse attention mechanism to overcome the limitations of fixed sparsity levels under diverse learner interaction patterns. By adaptively adjusting the sparsity according to the current interaction characteristics, the mechanism preserves computational linearity while more effectively selecting critical historical information and suppressing noise, thereby improving the stability and robustness of knowledge state estimation. The main contributions of this paper are as follows:

- We propose an efficient KT model named MCSKT, which significantly improves training and inference efficiency while maintaining strong prediction performance, offering a new perspective for knowledge modeling in large-scale online learning environments.
- We design a dual-encoder architecture based on Mamba to model the rich contextual information in sequences. Compared to attention-based contextual modeling method, our approach reduces computational complexity to linear, significantly improves the model's capability in processing long-sequence data.
- We design a dynamic k-sparse attention mechanism that adaptively filters key historical interactions, enabling the model to focus on information most relevant to the current question. By reducing the computation of irrelevant questions, this mechanism improves both model efficiency and prediction accuracy.
- Experimental results demonstrate that MCSKT outperforms comparison baselines in prediction accuracy, while significantly reducing time requirements, and thus achieving a dual improvement in prediction performance and efficiency. Furthermore, MCSKT exhibits superior computational efficiency compared to other methods as sequence length increases, demonstrating its advantages in handling long-sequence data.

The rest of the paper is organized as follows: Section 2 provides a comprehensive review of related works pertinent to the research problem. Section 3 presents the overall framework of MCSKT and its key components. Section 4 presents and analyzes the experiment. Section 5 discusses the limitations of the proposed model. Section 6 summarizes this paper and outlines directions for future research.

2. Related work

2.1. Deep learning-based knowledge tracing

Deep learning has garnered extensive attention from researchers due to its powerful feature extraction capabilities, bringing new opportunities and challenges to the field of KT (Sun et al., 2024). An increasing number of studies have applied deep learning techniques to KT task, making DLKT methods gradually become a research hotspot (Huang et al., 2024b; Gong et al., 2020). In 2015, Piech et al. were the first to apply deep learning techniques to knowledge tracing, and proposed the DKT based on RNN (Piech et al., 2015). This model transforms the learner's historical answering records into a time series and uses RNN to simulate the changes in the learner's knowledge state. However, DKT only uses the knowledge concepts covered by the questions and the answers as input, failing to fully exploit the characteristics of the learner's problem-solving process. Therefore, Su et al. proposed the EERNN (Su et al., 2018), which combines the learner's answering records with the text information of questions. Based on DKT, this model incorporates the text content of the exercises as an additional feature input to the Long Short-Term Memory (LSTM) network. Since learners exhibit forgetting behaviors during the learning process, Nagatani et al. proposed DKT-F (Nagatani et al., 2019), which adds a forgetting mechanism to DKT, enabling the model to better simulate the process of knowledge decay in learners. Shen et al. emphasized the importance of focusing on the consistency of changes in the learner's knowledge state and learning process, leading to the development of the LPKT (Shen et al., 2021). This model deeply measures learning gains and their diversity, which uses learning gates and forgetting gates to separately distinguish the learner's ability to absorb knowledge and simulate knowledge decay over time (Shen et al., 2021). However, the sequential dependency of RNNs leads to low training efficiency and makes it difficult to effectively capture long-range dependencies, and thus limiting their application in large-scale long-sequence scenarios.

In recent years, Transformer models have shown great potential in fields such as natural language processing, computer vision, and speech processing, and attention-based knowledge tracing models have gained widespread attention and application (Wang et al., 2025b, 2024, 2023a). By introducing the attention mechanism, these models can better handle long-range dependencies in sequences and improve

Table 1
Symbols and explanations.

Symbols	Explanations
q, c, r	Question, knowledge concept and response, respectively
x, y	Embedding of question and question–response, respectively
f	Embedding of forgetting feature
$\tilde{y}$	Forgetting-enhanced embedding of question–response
$\hat{x}$	Context-aware embedding of question
$\hat{y}$	Context-aware embedding of knowledge
k	Sparsity level of the learning interaction sequence
h_t	Knowledge state at time step t
$r_{t+1}, \hat{r}_{t+1}$	The actual response and predicted probability at time step $t+1$
L	Learning interaction sequence of the learner
Q, K, V	Query, Key and Value for the attention mechanism

parallelization during computation, significantly enhancing model prediction accuracy. Pandey et al. were the first to introduce the attention mechanism into the KT field, and proposed SAKT (Pandey and Karypis, 2019), which captures the relationships between the current exercise and historical interactions. Ghosh et al. proposed the AKT (Ghosh et al., 2020), which focuses on contextual information and uses a novel monotonic attention mechanism to assess the relationship between the current question and all previous questions. DTransformer (Yin et al., 2023) constructs the model from the question level to the knowledge level, which explicitly diagnose the learner's knowledge acquisition status, and uses a contrastive learning-based training algorithm to produce more stable tracing results. As attention mechanisms continue to be applied in knowledge tracing, sparse attention mechanisms have gradually gained attention due to their ability to optimize traditional attention mechanisms by computing only a limited number of attention scores in the sequence, rather than all possible pairs. These mechanisms have shown good performance across various fields (Chen et al., 2023). Huang et al. considered the issue of overfitting caused by data sparsity, and borrowed the idea of top-k mechanisms to propose two sparsification heuristics, thereby significantly improving the robustness and generalization ability of attention-based DLKT methods (Huang et al., 2023a).

2.2. Efficient knowledge tracing

Compared to early approaches (e.g., Bayesian Knowledge Tracing (BKT) (Corbett and Anderson, 1994b)), DLKT models are more applicable to handle large-scale data. However, they primarily focus on improving model prediction accuracy without ensuring high efficiency during training and inference. Selective state-space models effectively solve the long-term dependency problem and achieve linear complexity, which show unique advantages in handling long sequence data. Therefore, they have been widely applied in time series analysis, vision, and other fields (Behrouz and Hashemi, 2024). Inspired by this, Cao et al. creatively introduced Mamba into the KT field and proposed the Mamba4KT (Cao and Zhang, 2024), which combines Mamba with advanced technologies in traditional KT models. This model not only ensures prediction accuracy and interpretability but also achieves a balance between time and space consumption. However, the model still lags behind existing DLKT models in terms of prediction accuracy. Therefore, how to improve the model's training and inference efficiency while ensuring prediction performance remains a critical issue that needs to be addressed in the current study of knowledge tracing.

3. Method

In this section, we provide a detailed description of the proposed MCSKT method. Table 1 contains a list of symbols used in this paper.

3.1. Model overview

To address the challenge of balancing prediction accuracy with computational efficiency in existing KT models, we propose the MCSKT, whose overall architecture is illustrated in Fig. 2.

The MCSKT model consists of four modules: the two-stage feature embedding module, the context-aware representation module, the knowledge state extraction module, and the prediction module. Firstly, the two-stage feature embedding module uses Rasch model-based embedding method in the first stage to generate question embeddings and question–response embeddings by considering the individual differences between questions covering the same concept. Because learners exhibit some degree of forgetting during the learning process, the second stage of the embedding incorporates three types of learner's forgetting features to enhance the question–response embeddings. Secondly, the context-aware representation module consists of a dual-encoder built on Mamba, namely the question encoder (Q-encoder) and the knowledge encoder (K-encoder), which are used to perform context-aware modeling on question embeddings and question–response embeddings, thereby fully capturing the learner's response history and sequence dependencies. Thirdly, to optimize the process of constructing knowledge states, the knowledge state extraction module employs a dynamic k-sparse attention mechanism to retrieve the most relevant past knowledge for the current question, and thus filtering out redundant noise and ultimately obtaining the learner's current knowledge state. Finally, the prediction module uses the acquired knowledge state to predict the learner's performance on the next question.

3.2. Two-stage feature embedding module

To mitigate data sparsity, traditional KT methods typically index questions by their associated concepts, i.e., setting $q_t = k_t$ (Wang et al., 2025b). However, this approach treats questions covering the same knowledge concept as identical, thereby ignoring individual differences among these questions. To address this limitation, we introduce a Rasch model-based embedding technique during the first stage of feature embedding (Ghosh et al., 2020). Using this method, we construct a question embedding sequence x and a question–response embedding sequence y based on the learner's learning history. The formulation is given in Eq. (1):

$$\begin{aligned} x_t &= \mathrm{c}_{k_t} + \mu_{q_t} \cdot \mathrm{d}_{k_t} \\ y_t &= \mathrm{e}_{(k_t, r_t)} + \mu_{q_t} \cdot \mathrm{f}_{(k_t, r_t)} \end{aligned} \tag{1}$$

Where c_{k_t} denotes the embedding of the knowledge concept this question covers, $\mathrm{e}_{(k_t, r_t)}$ denotes the embedding of the concept–response pair, μ_{q_t} is a scalar difficulty parameter that controls how far this question deviates from the concept it covers. d_{k_t} is a vector that summarizes the variation in questions covering this concept, $\mathrm{f}_{(k_t, r_t)}$ is a vector that summarizes the variation among different concept–response pairs (Ghosh et al., 2020).

According to Eq. (1), x is generated by combining the concept embedding with the difficulty parameter of the question, thereby distinguishing different questions that share the same concept. On the other hand, y incorporates both the learner's response to a specific question and the underlying concept, reflecting the learner's performance on that concept. If the learner answers correctly, the model infers that the concept is mastered; otherwise, it is deemed unmastered. Thus, y can be interpreted as an indicator of the learner's conceptual mastery. However, using only the learner's response as the representation of mastery is insufficient, because learners often exhibit forgetting behaviors in real-world learning environments (Zhao et al., 2023), and their proficiency may decline over time due to forgetting factors (Huang et al., 2020). To address this problem, we incorporate three types of forgetting features in the second stage of embedding to enhance y: (1) repeated time gap: the time gap since the last attempt on a question associated with the same knowledge concept. (2) past trial counts: the

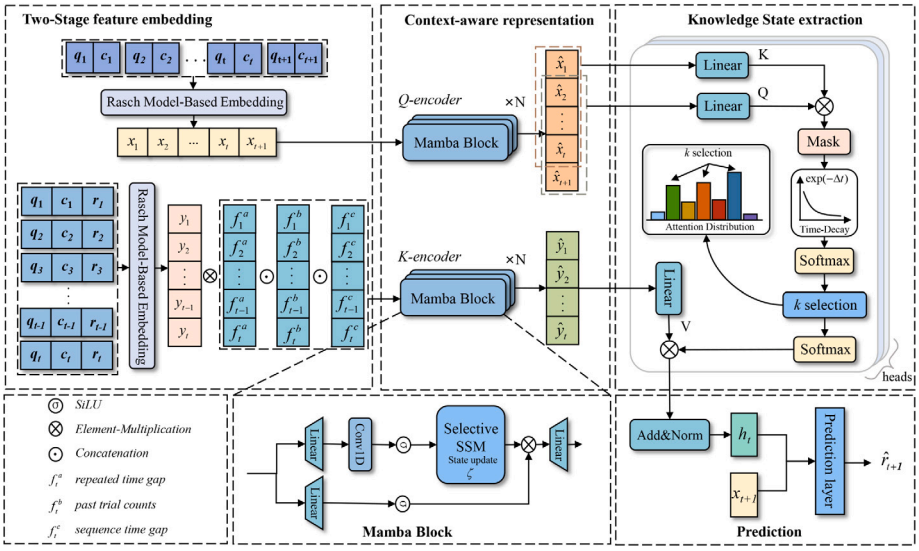


Fig. 2. The proposed MCSKT framework.

number of previous attempts on questions related to the same concept. (3) sequence time gap: the time gap since the last attempted question, regardless of the concepts involved (Nagatani et al., 2019). Specifically, we concatenate the three forgetting feature vectors into a single feature vector f, and fuse it with the question–response embedding y. The resulting enhanced embedding is denoted as $\tilde{y}$. The fusion follows the optimal method identified in Su et al. (2018), as shown in Eq. (2):

$$\tilde{y}_t = [y_t \otimes F_{f_t};\ f_t] \tag{2}$$

where F is a trainable transformation matrix, $\otimes$ denotes element-wise multiplication. Finally, this embedding module yields the question embedding sequence x and the enhanced question–response embedding sequence $\tilde{y}$.

3.3. Context-aware representation module

The evolution of a learner's knowledge state is closely related to their historical response sequence. Even the same question, learners with different response histories may exhibit distinct levels of understanding and mastery (Ghosh et al., 2020). However, raw embeddings often fail to fully capture the learner's historical context and response patterns. To address this problem, we propose a context-aware representation module, which incorporates a dual-encoder architecture based on the Mamba framework. This module includes a question encoder (Q-encoder) and a knowledge encoder (K-encoder), which are designed to learn contextual representations of questions and responses. Compared to raw embeddings, the representations generated by this method better reflect learners' true comprehension and the knowledge acquired through their personal response histories (Ghosh et al., 2020). Specifically, Mamba (Gu and Dao, 2023) employs a selective state space mechanism to dynamically filter long-range dependencies while maintaining linear time complexity. Through a gating mechanism, it discerns context relevance among sequence elements, and its hardware-aware parallel scan algorithm further accelerates training. Compared to RNN and attention mechanisms, Mamba achieves a superior trade-off between computational efficiency and long-sequence modeling capability. In practice, the Q-encoder takes the question embedding sequence $\{x_1, x_2, \ldots, x_t\}$ obtained from the embedding module as input. It outputs a series of context-aware question embeddings $\{\hat{x}_1, \hat{x}_2, \ldots, \hat{x}_t\}$, where each embedding is conditioned on the current question and its preceding questions. Similarly, the K-encoder takes the enhanced question–response embedding sequence $\{\tilde{y}_1, \tilde{y}_2, \ldots, \tilde{y}_t\}$ as input and outputs a sequence of actual knowledge representations $\{\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_t\}$ where each representation is influenced by the learner's responses to both the current and past questions (Wang et al., 2025b).

Mamba Block. The structure of the Mamba block is illustrated in Fig. 2. Its core design employs a dual-branch feature fusion mechanism to jointly model local and global contextual information. Specifically, the input is first linearly projected into two parallel paths: the first path passes through a one-dimensional convolution layer to extract local sequential features, followed by a SiLU activation function to enhance non-linear expressiveness. The result is then fed into the Selective State Space model (S6) (Gu and Dao, 2023), which reinforces and filters relevant long-range dependencies to model contextual information more effectively; the second path bypasses the convolution and directly applies SiLU activation, which aims at capturing global non-linear characteristics. After processing, the features from both branches are fused and passed through another linear projection to generate the final output vector, which serves as the context-aware embedding for either a question or a knowledge representation. Assuming the input is a question embedding $x = \{x_1, x_2, \ldots, x_t\}$, the output $\hat{x}$ represents the corresponding context-aware embedding vector, which is defined in Eq. (3):

$$\begin{aligned} m &= \mathrm{S6}(\mathrm{SiLU}(\mathrm{Conv1D}(\mathrm{Linear}(x)))) \\ n &= \mathrm{SiLU}(\mathrm{Linear}(x)) \\ \hat{x} &= \mathrm{Linear}(m \otimes n) \end{aligned} \tag{3}$$

3.4. Knowledge state extraction module

To obtain the learner's knowledge state at the current time step for subsequent prediction task, we propose a knowledge state extraction module that leverages an attention mechanism to retrieve relevant knowledge related to the current question from the knowledge acquired in the past, which is then treated as the learner's current knowledge state (Ghosh et al., 2020). However, traditional attention mechanism considers the attention weights between all pairs of questions, resulting in a time complexity of $\mathcal{O}(N^2)$, which not only reduces computational efficiency but also introduces noise from irrelevant information. To address this issue, we design a dynamic k-sparse attention mechanism, which introduces a learnable top-k selection operator to adaptively retain the most critical self-attention values. This allows the model to focus exclusively on information that is most relevant to the current question, thereby filtering out redundant noise and improving efficiency by avoiding computation over unrelated questions. Specifically, this module takes as input the context-aware question embeddings

$\hat{x}_{1:t}$ and question–response embeddings $\hat{y}_{1:t}$, and applies a multi-head sparse attention mechanism to extract the knowledge most pertinent to the current question. The output is the learner's knowledge state representation h_t at the current time step t. In the following section, we detail the core idea behind the dynamic k-sparse attention mechanism and explain the process by which the final knowledge state h_t is computed.

Dynamic k-Sparse Attention. The intuition behind this attention mechanism is illustrated in Fig. 2. Based on the original scaled dot-product attention, we introduce a time-decay component and a top-k selection module, enabling the model to account for the temporal influence of memory while adaptively retaining only the most critical self-attention values. This design effectively filters out irrelevant interactions, thereby incorporating temporal effects in the learning process and improving computational efficiency. Firstly, the time-decay module is inspired by the monotonic attention mechanism (Ghosh et al., 2020), which introduces a multiplicative exponential decay term into the standard self-attention function. The attention function is defined in Eq. (4):

$$\begin{aligned} &\alpha_{t,\tau} = \text{softmax}\left(\frac{Q_t^\top K_\tau}{\sqrt{d_k}} \cdot \exp(-\theta \cdot d(t-\tau))\right) \\ &\text{where } Q_t = \hat{x}_t \cdot W^Q, \quad K_\tau = \hat{x}_\tau \cdot W^K \end{aligned} \tag{4}$$

where $\theta > 0$ is a learnable decay-rate parameter and $d(t,\tau)$ denotes the temporal distance between steps t and τ. With this formulation, the attention weight assigned to past interactions depends not only on the similarity between the corresponding query and key vectors, but also on their relative temporal distance. As the distance increases, the temporal influence decays exponentially, meaning that earlier learning experiences have gradually diminishing impact on the current knowledge state (Yin et al., 2023). However, the quadratic computational complexity of standard attention grows rapidly as the number of historical interactions increases, severely reducing computational efficiency. Moreover, assigning attention weights to every past interaction may introduce noise, which can negatively affect the accuracy of knowledge state estimation (Huang et al., 2023a). To address these issues, we incorporate a learnable top-k selection operator on top of the time-decayed attention scores, ensuring that the knowledge state estimation focuses on only a limited number of the most influential interactions (Huang et al., 2023a; Amini and Ghaemmaghami, 2022). Let $A = \{\alpha_{t,1},\ \alpha_{t,2},\ \ldots,\ \alpha_{t,t-1}\}$ denote the vector of attention scores computed after the time-decay module. Let $M_{\text{topK}}(\cdot)$ be a masking operator that selects the top-k most influential past interactions. The parameter k serves as a tunable factor that dynamically controls the sparsity of the attention distribution. In practice, we apply normalization only to the top-k values of A that fall within the interval $[\Delta 1,\ \Delta 2]$, and these values then participate in the Softmax computation. Suppose s denotes the kth largest value in A, If $\alpha_{t,\tau} \geq s$, it is selected; otherwise, it is discarded. The specific operation is shown in Eq. (5):

$$M_{\text{topK}}(\alpha_{t,\tau}) = \begin{cases} \alpha_{t,\tau} & \text{if } \alpha_{t,\tau} \geq s \\ -\infty & \text{otherwise} \end{cases} \tag{5}$$

After applying the Top-K selection, we obtain a new attention score distribution $\hat{A} = \{\hat{\alpha}_{t,1},\ \hat{\alpha}_{t,2},\ \ldots,\ \hat{\alpha}_{t,t-1}\}$. Then $\hat{A}$ is re-normalized, and the normalized scores of $\hat{\alpha}_{t,\tau}$ in negative infinity are approximately 0. As a result, a sparse attention score distribution is obtained, which is subsequently used to compute a weighted sum over the corresponding value. Finally, the knowledge state representation of the ith attention head at time step t is yielded, as defined in Eq. (6):

$$\begin{aligned} &h_t^i = \sum_{\tau=1}^{t-1} \hat{\alpha}_{t,\tau} \cdot V_\tau \\ &\text{where } V_\tau = \hat{y}_\tau \cdot W^V \end{aligned} \tag{6}$$

The processing pipeline of the dynamic k-sparse attention mechanism is illustrated in Algorithm 1.

Furthermore, to further elucidate the theoretical rigor of the proposed mechanism, we perform a formal analysis of its performance gain

Algorithm 1 Dynamic k-Sparse Attention Mechanism.

Input: $\hat{x} = (\hat{x}_1, \ldots, \hat{x}_{t+1})$, $\hat{y} = (\hat{y}_1, \ldots, \hat{y}_t)$, sparsity function $k(\cdot)$
Output: h_t

$Q \leftarrow \text{Linear}_Q(\hat{x}_{t+1})$
for $i = 1$ **to** t **do**
  $K_i \leftarrow \text{Linear}_K(\hat{x}_i), \quad V_i \leftarrow \text{Linear}_V(\hat{y}_i)$
end for
$K = [K_1, \ldots, K_t], \quad V = [V_1, \ldots, V_t]$
for $i = 1$ **to** t **do**
  $S_i \leftarrow \frac{QK_i^\top}{\sqrt{d}}$
end for
$S \leftarrow \text{Mask}(S)$
// Time decay
for $i = 1$ **to** t **do**
  $\text{decay}_i \leftarrow \exp(-\theta \cdot d(t-i))$
  $A_i \leftarrow S_i \cdot \text{decay}_i$
end for
$A \leftarrow \text{Softmax}(A)$
// Dynamic k-sparse selection
$k^* \leftarrow k(A)$, $\text{idx} \leftarrow \text{TopK}(A, k^*)$
$\hat{A} \leftarrow 0$
$\hat{A}[\text{idx}] \leftarrow A[\text{idx}]$
$\hat{A} \leftarrow \text{Softmax}(\hat{A})$
$h_t \leftarrow \sum_{i=1}^{t} \hat{A}_i \cdot V_i$
return h_t

from the perspectives of generalization theory and gradient stability. Theoretically, by restricting the effective interaction space to k key items, the mechanism essentially performs implicit structural regularization on the attention operator, thereby effectively compressing the capacity of the model's effective hypothesis space. Specifically, according to statistical learning theory, the Rademacher complexity $R_n(\mathcal{H})$ of the MCSKT hypothesis class $\mathcal{H}$ is directly controlled by the sparsity k, with its theoretical bound significantly tightened to $R_n(\mathcal{H}) \leq \mathcal{O}(k/\sqrt{n})$. This reduction in complexity leads directly to a tighter generalization error bound:

$$E[f] \leq \hat{E}[f] + \mathcal{O}(R_n(\mathcal{H})) \tag{7}$$

where $E[f]$ and $\hat{E}[f]$ represent the expected error and empirical error, respectively. During the optimization process, since the gradients of low-relevance interactions are strictly masked to zero during computation, the model constructs a selective gradient path. This mechanism effectively minimizes the gradient variance $Var(\nabla_\theta \mathcal{L})$ during back-propagation, avoiding stochastic noise interference introduced by a large number of low-weight items, thereby enhancing the convergence stability of the training process. Detailed derivations are provided in Appendix.

Multi-head Attention. To enhance robustness and capture diverse aspects of the learner's interaction history during aggregation, the sparse attention mechanism is extended to a multi-head setting, in which H independent attention heads are employed (Yin et al., 2023). Each head is assigned its own decay rate θ and independently performs the full sparsification process described above. The outputs of all heads are concatenated and passed through layer normalization, and thus yielding the final knowledge state representation.

3.5. Prediction module

In the prediction module, the knowledge state h_t obtained from the knowledge state extraction module is concatenated with the embedding

x_{t+1} of the next question. This combined representation is passed through a two-layer fully connected network, followed by a sigmoid activation function to produce a prediction probability $\hat{r}_{t+1} \in [0,1]$, which represents the likelihood that the learner will answer the next question correctly. The computation is defined in Eq. (8):

$$\begin{aligned} f_1^t &= \mathrm{ReLU}(W_1(h_t \oplus x_{t+1})^\top + b_1) \\ f_2^t &= \mathrm{ReLU}(W_2 f_1^t + b_2) \\ \hat{r}_{t+1} &= \sigma(W_3 f_2^t + b_3) \end{aligned} \tag{8}$$

Where W_1, W_2, W_3, b_1, b_2 and b_3 are trainable parameters, and $W_1 \in \mathbb{R}^{d\times 2d}$, $W_2 \in \mathbb{R}^{d\times d}$, $W_3 \in \mathbb{R}^{1\times d}$, $b_1 \in \mathbb{R}^{d\times 1}$, $b_2 \in \mathbb{R}^{d\times 1}$, $b_3 \in \mathbb{R}^{1\times 1}$ (b_3 is a scalar), $\sigma(\cdot)$ is the sigmoid activation function.

To optimize all parameters in the model, we adopt the cross-entropy loss function to measure the discrepancy between the predicted response $\hat{r}_{t+1}$ and the ground-truth label r_{t+1}. To this end, we employ the Adam optimizer, and the objective function is computed as shown in Eq. (9):

$$L = -\sum_i \sum_t \left(r_{t+1} \log \hat{r}_{t+1} + (1 - r_{t+1}) \log(1 - \hat{r}_{t+1}) \right) \tag{9}$$

3.6. Summary of design mechanisms and methodologies

This section provides a comprehensive synthesis of the design rationale and methodology of the aforementioned MCSKT framework, elucidating its distinct advantages in complex KT tasks.

Firstly, in terms of decoupled modeling at the representation layer, unlike general sequence models that homogenize input signals, MCSKT constructs a dual-encoder architecture (Q-encoder and K-encoder) specifically designed for the characteristics of educational data. This architecture fully considers the hierarchical differences between question semantics and knowledge mastery in knowledge tracing tasks. This design not only decouples the representation of content semantics and cognitive states in structure but also helps the model accurately recognize the dynamic evolution of learners' knowledge states in long-term interaction sequences, thereby significantly enhancing the model's feature representation capability in complex educational scenarios. Secondly, to balance computational efficiency with long-term memory, the dynamic k-sparse attention mechanism serves as a critical mechanistic supplement to Mamba's linear modeling capability. Rather than relying on static empirical thresholds, this mechanism adaptively filters key historical interactions based on the learner's current real-time state. Inspired by the concept of frequency-domain filtering in recent spectral graph neural networks (Ouyang et al., 2025), this design can be interpreted as context-adaptive signal filtering in the time domain. Specifically, the exponential decay term functions as a low-pass filter that smooths out instantaneous random fluctuations to extract the learner's long-term and stable knowledge mastery trends. Meanwhile, the dynamic top-k selection acts as an adaptive band-pass filter, suppressing redundant noise to precisely capture critical historical interactions. Theoretically, this adaptive design optimizes the computational complexity of attention from $O(L^2)$ to $O(kL)$, achieving a dynamic balance between noise suppression and key information retention.

In summary, the architectural design of MCSKT is far more than a mere stacking of modules. Through structural representation disentanglement and adaptive mechanical optimization, it establishes a robust balance between algorithmic efficiency and predictive accuracy. This provides a solid theoretical foundation for real-time knowledge tracing in large-scale online education scenarios.

4. Experiments

To comprehensively evaluate the performance and effectiveness of the proposed MCSKT model, we conducted extensive experiments on four real-world datasets. The main goals of our experimental study include the following four aspects:

(1) Performance and efficiency evaluation: comparing MCSKT with existing KT models in terms of prediction accuracy and computational efficiency to validate its overall advantages;

(2) Module effectiveness analysis: evaluating the contribution of key components to prediction performance through ablation studies;

(3) Sequence length sensitivity analysis: investigating how different sequence lengths affect the efficiency and performance of MCSKT, thereby verifying its stability in long-sequence learning scenarios;

(4) Impact of sparsity level: examining the trade-off between computational efficiency and accuracy under different sparsity ratios.

In this section, we first provide detailed descriptions of the datasets, baseline models, evaluation metrics, and experimental settings. Subsequently, we conduct comparative experiments to evaluate the model's prediction performance and computational efficiency, along with ablation studies on key components. Additionally, supplementary experiments analyze the effects of varying sequence lengths and sparsity levels on the model. Finally, we present visualization analysis of the sparse attention mechanism's effectiveness and the evolution of learners' knowledge states.

4.1. Experimental setup

4.1.1. Datasets

We evaluate the proposed method and several baselines on four publicly available datasets in terms of both predicting learners' future performance and computational efficiency. Each dataset consists of real-world collected data. The statistics of all datasets are summarized in Table 2. The descriptions of the datasets are as follows:

- Assist12[1]: This dataset records learners' online interaction data from the 2012–2013 academic year on the online tutoring system named ASSISTments. Since many records in the raw data lack associated knowledge concepts, we filtered out those entries without concept annotations. Among the four selected datasets, Assist12 contains the largest number of learners, questions, and interaction records.
- Assist17[2]: This dataset is another release from ASSISTments, published as part of a long-term data mining competition. It includes over ten years of learner online interaction data on the ASSISTments platform. Among the four datasets, Assist17 has the longest average learning record length per learner.
- NIPS34[3]: This dataset is derived from Tasks 3 and 4 of the NeurIPS 2020 Education Challenge. The challenge provides answer records of mathematics questions collected between the 2018 and 2020 academic years on the Eedi platform, which is a leading educational platform used daily by millions of learners worldwide.
- EdNet[4]: This dataset originates from Santa, an AI-powered tutoring system, and contains learner interaction data collected by the system. Due to its large scale, we randomly sampled data from 5000 learners for our experiments.

4.1.2. Baselines

To comprehensively compare with the proposed model, we select nine models. Among them, DKT, DKT-Forget, LPKT, DIMKT and HD-KT are based on RNN structure; AKT, SparseKT, DTransformer and TCKT are attention-based models; Mamba4KT is the first knowledge tracing model leveraging the Mamba framework. The detailed descriptions of these models are as follows:

[1] https://sites.google.com/site/assistmentsdata/home/2012-13-school-data-with-affect

[2] https://sites.google.com/view/assistmentsdatamining/dataset

[3] https://eedi.com/projects/neurips-education-challenge

[4] https://github.com/riiid/ednet

Table 2
Statistics of datasets.

Datasets	#Learners	#Questions	#Concepts	#Interactions	#Avg.Length
Assist12	27 485	53 091	265	2, 709, 647	99
Assist17	1709	3162	102	942, 816	552
NIPS34	4918	948	57	1, 382, 727	282
EdNet	5000	12 060	189	580, 624	117

- DKT (Piech et al., 2015) is the first deep learning-based knowledge tracing model that employs LSTM to obtain hidden representations for tracking learners' knowledge states.
- DKT-F (Nagatani et al., 2019) is an extension of the DKT model that enhances prediction performance by incorporating multiple features derived from the learning sequence and forgetting behavior.
- AKT (Ghosh et al., 2020) combines a monotonic attention mechanism with Rasch model-based embeddings, which applies exponential decay on attention weights according to the temporal distance in the question sequence to simulate learner forgetting behavior.
- LPKT (Shen et al., 2021) directly models the learner's learning process to assess knowledge state, which employes a learning gate to control knowledge acquisition and a forgetting gate to determine the decay of knowledge over time.
- DIMKT (Shen et al., 2022) considers the significant impact of question difficulty on learning by using an adaptive sequence neural network (ASNN) to model the relationship between learner knowledge states and difficulty levels, and thus enhancing prediction accuracy.
- DTransformer (Yin et al., 2023) is the first architecture focusing on stable knowledge state estimation and tracking, rather than solely predicting future performance. It leverages Transformer models and a novel training paradigm with contrastive learning to maintain diagnostic stability of the knowledge state.
- SparseKT (Huang et al., 2023b) improves upon SAKT by employing two simple yet effective sparse attention heuristics to extract relevant information from a learner's past sequence, thereby improving prediction by filtering out noise from irrelevant historical interactions.
- Mamba4KT (Cao and Zhang, 2024) is the first to explore enhancing efficiency and resource utilization in knowledge tracing by introducing Mamba into the domain. It integrates advanced techniques from traditional KT models to improve predictive performance while ensuring time and space efficiency.
- TCKT (Huang et al., 2024a) is a KT method that integrates causal inference with temporal information, using front-door adjustment to build Causal Self-Attention for reducing data bias and employing a Learning Behavior Simulation module to model knowledge acquisition and forgetting.
- HD-KT (Ma et al., 2024) is a robust KT method that detects and removes anomalous learning interactions through dual anomaly detectors guided by knowledge states and student profiles, thereby improving the stability of knowledge state estimation and prediction accuracy.

4.1.3. Evaluation metrics

We use AUC (Area Under the ROC Curve) and ACC (Accuracy), which are common metrics in KT research, to evaluate the prediction performance of the models. Additionally, to comprehensively evaluate training and inference efficiency, we analyze several key metrics: the average running time per training epoch TT (in seconds), running time during testing IT (in seconds), number of model parameters (in MB), average FLOPs (in GFLOPs), and maximal GPU memory consumed during training (in GB) (Luo et al., 2024). To ensure fairness and prevent out-of-memory errors, the batch size for all models is fixed at 64 during both training and inference phases.

4.1.4. Experimental settings

For each dataset, 80% of learner data is used as the training set and 20% as the test set. During training, we perform standard 5-fold cross-validation for hyperparameter tuning; within each fold, the training set is further split into training and validation subsets at a 4:1 ratio. The two time gap features of forgetting features are calculated on a minute-by-minute basis, and all three forgetting features are discretized at $\log_2$ scale. In the experiments to investigate the effects of different sequence lengths (Section 4.4) and different sparsity levels (Section 4.5) on the models, the batch size is set to 24, while the batch size for the remaining experiments is fixed at 64. Furthermore, the maximum sequence length is set to 200 for all experiments except those in Section 4.4. For all methods, we utilize the Adam optimizer with a learning rate of 1e-5 to better control weight decay. To prevent overfitting, a dropout rate of 0.05 is applied. Training halts after 200 epochs or earlier if an early stopping criterion is met—specifically, training stops if the best AUC and ACC values do not improve for 20 consecutive epochs. All models are trained on a Linux server equipped with an NVIDIA RTX 3060 Ultra GPU.

4.2. Overall results

4.2.1. Comparison of effectiveness with baselines

To evaluate the effectiveness of the proposed model, we compare it with baseline models on four datasets. Table 3 summarizes the prediction performance of all methods, using AUC and ACC as evaluation metrics. As shown in the table, MCSKT outperforms all other methods across all datasets, achieving the highest scores on both AUC and ACC. On average, MCSKT improves AUC and ACC by 3.7% and 2.9%, respectively.

Compared with RNN-based models, our method achieves consistently higher AUC across all datasets. RNN-based approaches rely on sequential state updates, which hinder parallelization and suffer from gradient decay when modeling long sequences, making it difficult for them to capture long-range dependencies. By contrast, MCSKT replaces the traditional recurrent mechanism with Mamba's state-space architecture, enabling linear-time computation while preserving temporal consistency and effectively modeling long-term knowledge dependencies. Compared with attention-based models, although these models support parallel computation, their global attention design leads to quadratic computational complexity and increased sensitivity to noisy interactions, which significantly affects both efficiency and stability under long sequences and large-scale data. MCSKT mitigates these issues through linear state-space modeling combined with dynamic k-sparse attention. Furthermore, although Mamba4KT is the first to introduce a linear state-space mechanism into KT and improves computational efficiency, its architecture remains a single-encoder design and is not optimized for the semantic-level and cognition-level modeling requirements of KT tasks. Building upon this foundation, MCSKT adopts a dual-encoder architecture that separately models contextual information at the question level and the concept level, and incorporates dynamic sparse attention for adaptive feature selection. These structural improvements enhance the model's ability to capture complex learning behaviors, enabling MCSKT to outperform Mamba4KT across all four datasets.

Table 3
Overall results of performance prediction.

Model	Venue	Assist12		Assist17		NIPS34		EdNet	
		AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
DKT	NIPS'15	0.7303	0.7335	0.7171	0.6862	0.7712	0.7046	0.6505	0.6787
DKT-Forget	WWW'19	0.7333	0.7343	0.7258	0.6914	0.7675	0.7013	0.6653	0.6457
LPKT	KDD'21	0.7720	0.7527	0.7939	0.7412	0.8001	0.7287	0.7721	0.7300
DIMKT	SIGIR'21	0.7552	0.7453	0.7872	0.7272	0.8074	0.7471	0.7731	0.7096
AKT	KDD'20	0.7795	0.7598	0.7651	0.7243	0.8024	0.7326	0.7201	0.7088
SparseKT	SIGIR'23	0.7659	0.7422	0.7705	0.7170	0.7919	0.7227	0.7689	0.7053
DTransformer	WWW'23	0.7698	0.7519	0.7536	0.7078	0.7673	0.6842	0.7401	0.6944
Mamba4KT	arXiv'24	0.7764	0.7543	0.7818	0.7251	0.8033	0.7328	0.7526	0.7184
TCKT	ESWA'24	0.7763	0.7742	0.7965	0.7351	–	–	0.7671	0.7283
HD-KT	WWW'24	0.7797	0.7632	0.7714	0.7231	–	–	–	–
MCSKT (ours)	–	**0.8150**	**0.7816**	**0.8174**	**0.7604**	**0.8273**	**0.7635**	**0.7954**	**0.7472**

Table 4
Overall results of model efficiency.

Model	TT	IT	Parameter	FLOPs	GPU usage
DKT	17.53	2.25	0.37	4.82	0.37
DKT-Forget	19.20	2.43	0.44	5.77	0.76
LPKT	287.84	38.07	**0.07**	42.31	4.90
DIMKT	206.33	26.25	0.23	**3.76**	**0.36**
AKT	169.18	18.18	4.88	72.41	9.30
SparseKT	187.35	23.14	1.85	39.46	7.96
DTransformer	784.26	15.66	1.12	64.00	16.38
Mamba4KT	**39.52**	**4.2**	2.9	28.23	2.0
MCSKT	77.86	4.52	5.69	33.22	1.90

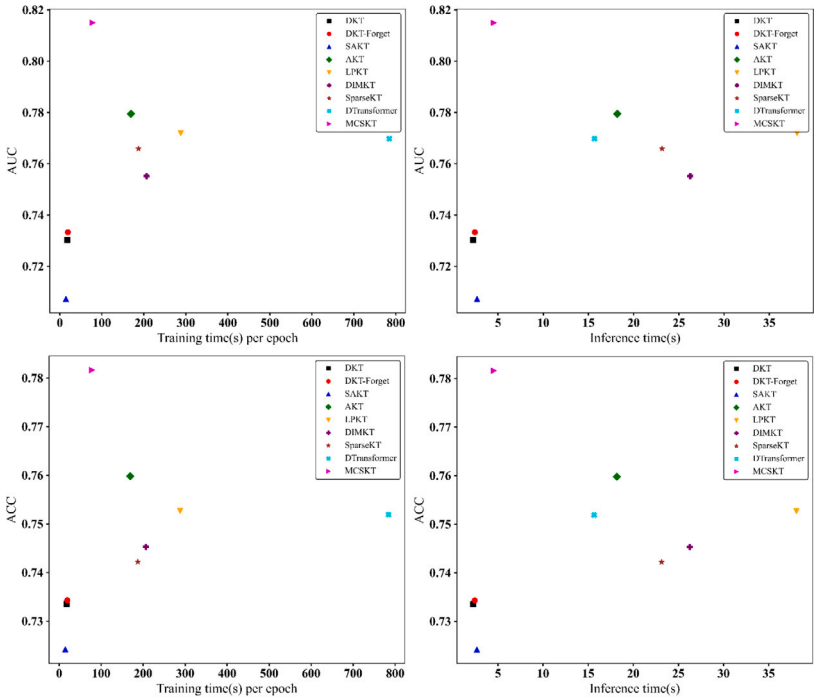


Fig. 3. Model comparison of the time efficiency on Assit2012.

4.2.2. Comparison of efficiency with baselines

To comprehensively evaluate the training and inference efficiency of the model, we compare MCSKT with several mainstream KT methods. To ensure fairness and prevent out-of-memory errors, all models use a fixed batch size of 64 and a sequence length of 200 during both training and inference. Table 4 presents the efficiency metrics of all models on the Assist12 dataset. The results indicate that our model requires only 77.86 s per epoch for training and 4.52 s for inference. Compared to the sequential model LPKT, our method achieves 3.6-fold and 8.4-fold efficiency improvements in training and inference efficiency, respectively. Furthermore, it even achieves a 10-folds reduction in training time over the DTransformer. Regarding resource consumption, MCSKT's GPU memory usage is only 11.6% of that of DTransformer, and its FLOPs are reduced by 54.1% compared to AKT. Notably, earlier models such as DKT, DKT-Forget, have shorter training times but underperform MCSKT in prediction accuracy, and their inference efficiency is not significantly superior to MCSKT, indicating the limitations of traditional approaches in the tradeoff of accuracy and speed. Fig. 3 provides a visual comparison of the proposed model against other baselines on the Assist2012 dataset in terms of runtime and prediction performance. The y-axis represents AUC or ACC, while

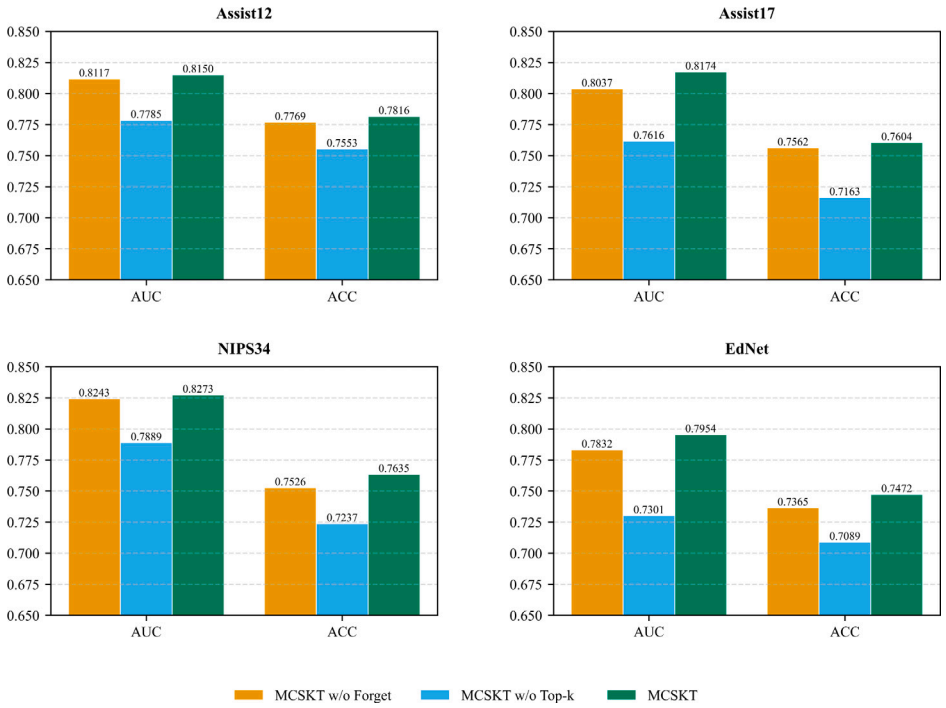


Fig. 4. Prediction Performance of the ablation variants across four datasets.

the x-axis denotes the average training time per epoch or the inference Time. Models positioned closer to the upper-left corner of the plot indicate better performance (Wang et al., 2025a). As shown in Fig. 3, our proposed model achieves the best prediction performance while maintaining acceptable runtime, clearly illustrating the improvements in both accuracy and efficiency offered by our approach.

Although Mamba4KT emphasizes lightweight architectural design, its primary objective is to improve training and inference efficiency rather than to enhance representational capacity through additional structural mechanisms. In contrast, MCSKT builds upon the Mamba architecture by independently encoding questions and knowledge concepts, enabling a more comprehensive representation of students' knowledge evolution and the deeper dependencies among questions. While these enhancements introduce moderate computational overhead, they substantially strengthen the model's sequence modeling capability, allowing MCSKT to achieve more expressive learning behavior characterization and knowledge state representation while maintaining high efficiency.

It is worth noting that the parameter size of MCSKT (5.69M) is relatively larger than some baselines. This is mainly because the model incorporates several enhanced components beyond the Mamba backbone: the Rasch embedding extension that captures item difficulty and concept relations, the dual-encoder structure that separately models question and response sequences, and the dynamic k-sparse attention module designed to accurately extract knowledge states. These components significantly improve the model's expressive power but also introduce additional learnable parameters. Nevertheless, thanks to Mamba's linear state-update mechanism, MCSKT continues to demonstrate strong performance in terms of training speed, inference efficiency, and memory consumption, achieving a well-balanced trade-off between effectiveness and efficiency.

Furthermore, the resource occupancy and efficiency metrics of MCSKT provide empirical indicators for its potential deployment across diverse hardware platforms, such as mobile terminals and edge AI modules. As illustrated in Table 4, MCSKT maintains a memory footprint of 1.9 GB, representing only 11.6% of that required by DTransformer. This reduced resource demand, coupled with 33.22 GFLOPs, suggests the feasibility of adapting the model to memory-constrained environments without a significant compromise in performance. Such efficiency is primarily attributed to the synergy between Mamba's linear state-update mechanism and the k-sparse weight-space pruning, which collectively alleviate the computational burden. By achieving a stable inference time of 4.52s, these empirical results lay a solid foundation for implementing real-time, cross-platform knowledge tracking in practical intelligent education scenarios.

4.3. Ablation study

To validate the effectiveness of the key modules in our proposed method, we conduct ablation experiments. Table 5 presents the ablation results across four datasets, and Fig. 4 provides a more intuitive visualization of performance.

Here, "w/o Forget" refers to the variant where forgetting features are excluded from the two-stage feature embedding module, while "w/o Top-K" denotes the knowledge state extraction module using monotonic attention mechanism that considers only temporal effects, with the Top-k selection module removed. The results in Table 5 reveals that removing forgetting features leads to a decrease in both AUC and ACC across all datasets, verifying the effectiveness of multi-dimensional forgetting features in modeling learners' knowledge decay. Disabling the dynamic Top-k selection module causes a significant performance drop, demonstrating that the sparse attention mechanism improves robustness by focusing on critical historical interactions and suppressing irrelevant noise. This further demonstrates the significant contribution of the proposed dynamic sparse attention mechanism in enhancing model performance.

4.4. Impact of sequence length on model efficiency

To deeply evaluate the impact of different sequence lengths on model computational efficiency, and particularly to verify the engineering advantages of the Mamba architecture in processing large-scale data, we conducted a comparative analysis and stress testing under extreme lengths between MCSKT and several mainstream knowledge tracing models. Fig. 5 illustrates the average training time per epoch and inference time of each model on the Assist2017 dataset under

Table 5
Results of the ablation study.

Method	Forget feature	Top-K	Assist12		Assist17		NIPS34		EdNet	
			AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
MCSKT w/o Forget	✗	✓	0.8117	0.7769	0.8037	0.7562	0.8243	0.7526	0.7832	0.7365
MCSKT w/o Top-k	✓	✗	0.7785	0.7553	0.7616	0.7163	0.7889	0.7237	0.7301	0.7089
MCSKT	✓	✓	**0.8150**	**0.7816**	**0.8174**	**0.7604**	**0.8273**	**0.7635**	**0.7954**	**0.7472**

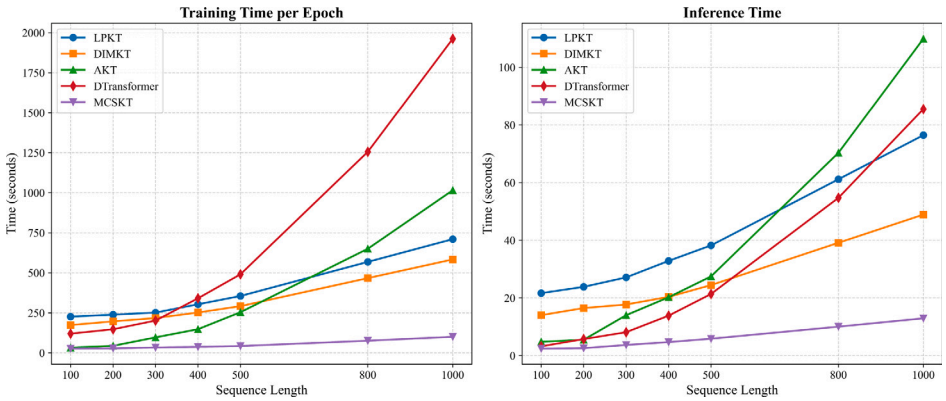


Fig. 5. Comparison of model time when dealing with sequences of different lengths.

sequence lengths $L \in \{100, 200, 300, 400, 500, 800, 1000\}$. Among the baselines, AKT and DTransformer are attention-based models, while LPKT and DIMKT are RNN-based models. Given that the average length of learning records in the Assist2017 dataset is approximately 500, the sequences of lengths 800 and 1000 were generated through oversampling and concatenating the original interaction sequences, aiming to simulate ultra-long temporal scenarios. To prevent out-of-memory issues, the batch size for all models was uniformly set to 24.

It can be observed from the results that as the sequence length increases, the computational overhead of each model exhibits distinctly different growth trends. Attention-based models (e.g., DTransformer and AKT) are constrained by the $O(L^2)$ computational complexity, with their training and inference times showing significant quadratic growth as the sequence length increases. Notably, during the stress testing phase ($L = 1000$), the time consumption of DTransformer increases dramatically. Meanwhile, although RNN-based models maintain a linear growth trend, their overall time consumption remains at a relatively high level due to the inherent bottleneck of sequential hidden state computation. In contrast, MCSKT maintains linear complexity thanks to Mamba's selective state space structure. In the extreme case of $L = 1000$, its training and inference speeds are significantly superior to those of Transformer-based models. This validates that MCSKT can effectively alleviate the computational burden brought by long sequences when processing ultra-long learning trajectories, demonstrating superior scalability for real-world engineering applications.

4.5. Impact of different sparsity levels on model performance

We further investigate how varying sparsity levels affect model performance and training time. Experiments were conducted on the Assist2017 dataset to evaluate the impact of different values of k on train set performance, with the results shown in Fig. 6.

From the results, it can be observed that manually setting k to a fixed value—such as half the sequence length—makes the model's performance highly sensitive to k and results in increased training time. To avoid exhaustive search, we instead define a controllable interval for k that allows dynamic learning of the most contributive values. We find that when the interval $[\Delta 1, \Delta 2]$ is set to [1/4, 2/3], the model achieves the highest prediction accuracy on the test set, with an ACC of 0.7604, which outperforms other intervals. Notably, although the interval [1/6, 1/3] yields the shortest training time, its accuracy is significantly lower than that of the [1/4, 2/3] interval. The training time for [1/6, 1/3] is 28.43 s per epoch—slightly higher than the minimum but far lower than that of higher intervals—indicating an optimal balance between accuracy and computational efficiency. Thus, we hypothesize that a smaller k, such as within the [1/6, 1/3] range, the selected historical interactions are fewer, and the KT model cannot acquire sufficient past learning information to predict learners' future performance, thereby leading to lower ACC. Expanding the sparse sampling range enables the model to capture richer past information, thus improving predictive accuracy. However, extreme sparsity, such as when k being 1/2 of the sequence length, or overly dense sampling, e.g., [1/2, 4/5], introduces more noise, reducing model robustness and limiting generalization capability, which leads to a decrease in ACC and computational efficiency.

4.6. Rationality analysis of the dual-encoder design

To validate the rationality and necessity of the dual-encoder architecture in MCSKT for decoupling question-level semantics and concept-level cognitive states, we conducted a feature-space correlation analysis. Specifically, we analyzed the correlation between the contextual question representation $\hat{x}$ generated by the Q-encoder and the knowledge state representation $\hat{y}$ generated by the K-encoder. To quantitatively evaluate the effectiveness of this disentanglement, we extracted the hidden state features of all interaction samples from the Assist2017 test set and calculated their Cosine Similarity.

Fig. 7 illustrates the distribution histogram of the similarity scores, where the vertical axis represents the frequency of interaction samples. We observed that the average cosine similarity between the two representations is 0.5128. This distribution pattern confirms that MCSKT successfully achieves structural disentanglement: the Q-encoder focuses on modeling static semantics and difficulty correlations among questions, while the K-encoder is dedicated to tracking the dynamic evolution of learners' knowledge mastery. Such a division of labor not only prevents interference from question-level background noise during the estimation of cognitive states but also ensures that the model can capture deeper, non-redundant information when processing ultra-long sequences of learning interactions.

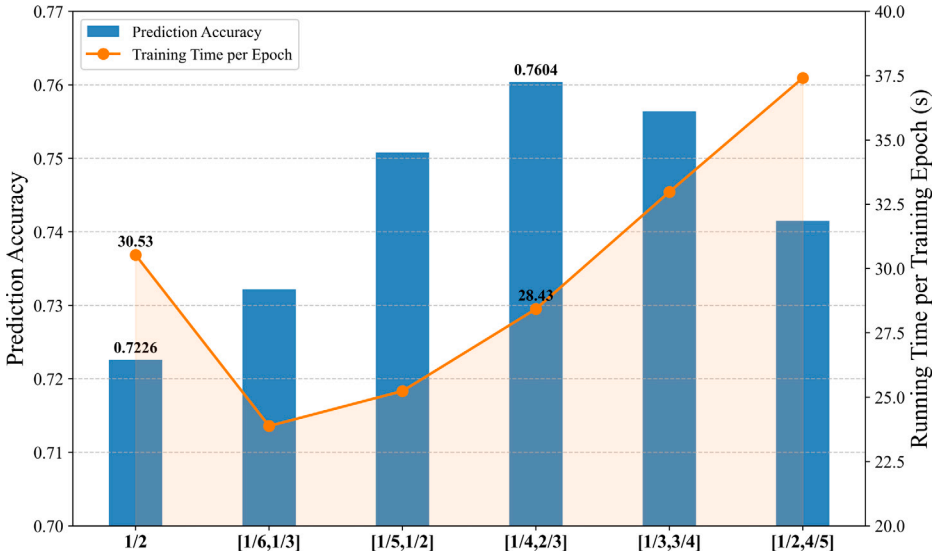


Fig. 6. Accuracy and Running Time per training epoch of different value of k on Assist17.

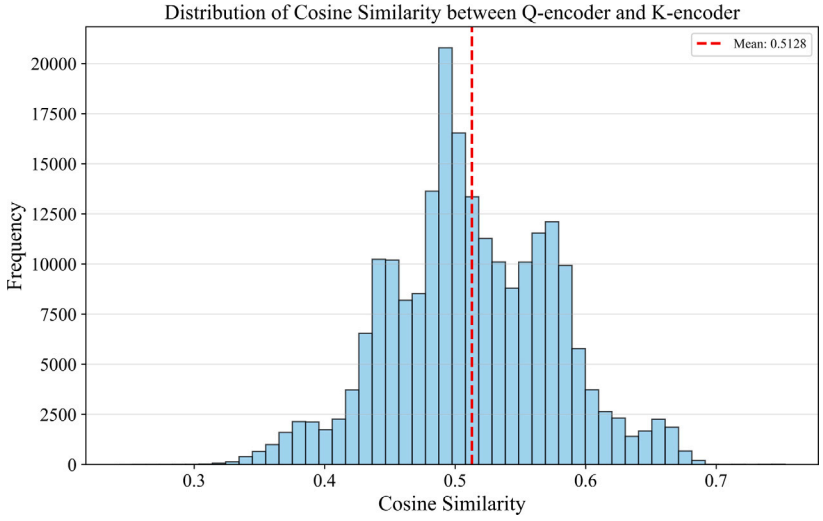


Fig. 7. Distribution of cosine similarity between output features of Q-encoder and K-encoder on the Assist17.

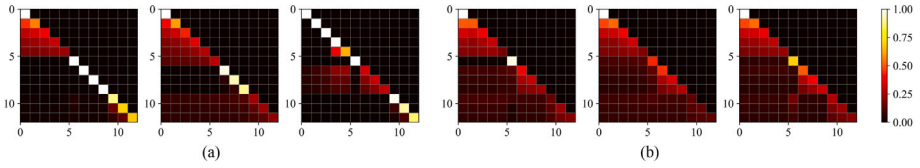


Fig. 8. Visualization of attention weights for different attention mechanisms. (a) Attention weights of the k-sparse attention mechanism; (b) Attention weights of the plain attention mechanism.

4.7. Visualizations

4.7.1. Visualization of the k-sparse attention mechanism

To demonstrate the advantages of our designed sparse attention mechanism in temporal decay and key interaction selection, we visualize attention weights using a learner's interaction sequence. The results are shown in Fig. 8.

Fig. 8 illustrates the attention weights over the 12 consecutive questions of a particular learner, with three attention heads randomly selected for visualization. Furthermore, we highlight the distinction between our sparse attention mechanism and the plain attention mechanism. As shown in Fig. 8(a), the attention weights in our proposed model are noticeably sparser than those in the plain attention mechanism. This indicates that our sparse attention mechanism not only identifies critical interactions but also effectively filters out irrelevant information, enabling the model to focus on the most important historical information relevant to the current question and thus improving prediction performance. In contrast, Fig. 8(b) shows that the plain attention mechanism focuses on all interactions indiscriminately, which not only introduces noise but also reduces inference efficiency. Additionally, we observe that each attention head operates at its own temporal scale, some focus primarily on recent interactions, while others attend to more distant history, further indicating the impact of temporal decay on model predictions.

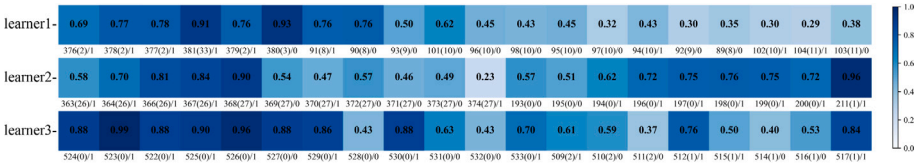


Fig. 9. Visualization of knowledge states.

4.7.2. Visualization of learners' knowledge states

Tracking the evolving knowledge state of learners as they continue practicing is a critical task in KT (Zhao et al., 2023). Therefore, this study employs the MCSKT model to analyze the knowledge tracing process of learners in the NIPS34 dataset. We randomly selected three learners and their response sequences for 20 questions. The correspondence between their responses and the predicted knowledge states is shown in Fig. 9. In each row, the indices represent the question id and the learner's response, with the knowledge concept covered by each question indicated in parentheses.

In the answer sequence for learner 1, questions involving the same knowledge concepts occur multiple times (e.g., concepts 2 and 10). When the learner answers concept 10 incorrectly twice consecutively, the corresponding knowledge state tends to decline. Conversely, the knowledge state for concept 2 significantly improves with consecutive correct responses. Notably, the predicted probability for question 5 is lower than that for question 4, indicating that the attention mechanism selectively focuses on more relevant interactions rather than relying solely on recent performances. For Learner 2, the response sequence frequently involves concepts 27 and 0. The knowledge state for concept 0 shows a generally smooth upward trend as the learner answers correctly in succession. In contrast, for concept 27, the learner's alternating correct and incorrect responses cause fluctuations in the corresponding knowledge state values. Finally, Learner 3's sequence contains 12 questions involving concept 0. Between questions 7 and 9, the predicted probability experiences a sharp drop followed by a rapid recovery, reflecting forgetting behavior that lowers the probability and dynamic selection of recent incorrect interactions as key cues. Subsequent rapid adjustment via the attention mechanism demonstrates the model's responsiveness to recent positive feedback. Experimental results shows that the predicted probabilities produced by our model closely align with the learners' actual responses, and thus validating the effectiveness of the model design.

5. Limitation and discussion

To address the challenge of balancing prediction accuracy and computational efficiency in current KT research, we propose an efficient KT model named MCSKT, which aims to achieve simultaneous improvements in both prediction performance and efficiency. By leveraging the linear computational complexity of the Mamba architecture and the adaptive filtering capability of the dynamic sparse attention mechanism, MCSKT significantly reduces computational costs while effectively enhancing prediction performance and model robustness. To evaluate the effectiveness of the proposed method, we conducted experiments on real-world datasets. Table 3 shows that MCSKT outperforms the second-best baseline by 3.7% in AUC and 2.9% in ACC, demonstrating superior predictive capability. In addition, to assess the efficiency of our method, we carried out extensive evaluations on various time-related metrics. As shown in Table 4, MCSKT requires less training and inference time compared to most existing KT models and also exhibits significantly lower FLOPs and memory consumption. Furthermore, we investigated the impact of varying sequence lengths on model efficiency. Experimental results indicate that as the sequence length increases, the growth in time consumption for MCSKT is substantially slower than that of competing models, which highlights its advantage in handling long sequential data. Despite achieving a favorable trade-off between accuracy and efficiency, MCSKT still has limitations. Specifically, it does not explicitly model the semantic relationships between knowledge concepts (Dai et al., 2024), which may constrain the model's ability to capture and interpret student knowledge states at a fine-grained level.

6. Conclusion

In this paper, we address the fundamental challenge in KT of balancing prediction accuracy and computational efficiency by proposing an efficient model named MCSKT, which integrates Mamba-based contextual encoding with a dynamic sparse attention mechanism. Leveraging the linear computational complexity of the selective state space model, MCSKT achieves efficient contextual modeling of long-sequence learner behaviors in KT. Furthermore, the designed dynamic k-sparse attention mechanism adaptively selects the most relevant historical interactions for the current question, effectively suppressing irrelevant noise and enhancing model robustness. Extensive experiments on multiple real-world datasets demonstrate that MCSKT outperforms the counterparts in prediction accuracy while substantially reducing computational time, thereby achieving a dual improvement in performance and efficiency.

In the future, we will focus on addressing the current model's limitation in explicitly modeling the semantic relationships between knowledge concepts. We plan to enhance Mamba's Selective State Space Model by incorporating Knowledge Graphs (KG) or concept hierarchies. Specifically, we intend to utilize semantic proximity to impose regularization constraints on the state transition matrix, enabling the precise propagation of learning gains across semantically related knowledge points and thereby providing the model with more fine-grained diagnostic capabilities. Additionally, we will further validate the model's generalization ability on larger-scale and more diverse educational datasets, while exploring the integration of multimodal learning behavior data to construct a more comprehensive and interpretable framework for knowledge state modeling.

CRediT authorship contribution statement

RuiJuan Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Feng Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Cong Liu:** Writing – review & editing, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.engappai.2026.114312.

Data availability

The data used in this study are publicly available.

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